

Problem Statement

Extractive distillation is widely used for ethanol dehydration because conventional distillation cannot separate the ethanol-water azeotrope.

Although recent studies have demonstrated the feasibility of modelling this process in DWSIM, published literature generally provides only sensitivity plots and final conclusions rather than the complete simulation specifications required for process reproduction.

During this work, several practical challenges were encountered:

- Missing operating specifications prevented direct replication of the published process.
- Multiple assumptions had to be developed through engineering judgment.
- Column convergence depended on the interaction of several operating variables rather than a single parameter.
- Hundreds of trial-and-error simulations were required to identify feasible operating regions.
- Manual optimization became increasingly time-consuming as more variables were considered simultaneously.

To address these limitations, this project proposes an engineering-guided digital twin framework that combines first-principles process simulation with machine learning.

The methodology first establishes a validated ChemSep simulation model, performs systematic sensitivity analysis to determine feasible operating ranges, generates an engineering dataset from converged simulations, and finally trains machine learning models capable of predicting overall process performance.

An AI-guided optimizer is then used to recommend promising operating conditions, allowing engineers to reduce simulation effort while maintaining physically meaningful process recommendations.